
Review

Reviewed Work(s): Measurement, Design and Analysis: An Integrated Approach. by E. J. Pedhazur and L. Pedhazur Schmelkin

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Source: *Journal of the Royal Statistical Society. Series D (The Statistician)*, 1994, Vol. 43, No. 4 (1994), pp. 608-609

Published by: Wiley for the Royal Statistical Society

Stable URL: <https://www.jstor.org/stable/2348155>

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will not need to wade through extra books to understand the techniques explained. Having said that, however, every chapter is supplemented with a bibliography section presenting a very useful review of references in the field. This review points readers into the direction of more complicated survival analysis, although it would be nice to see such analysis in a future edition, perhaps, for example, crossover designs.

This second edition contains plenty of new material such as the analysis of prognostic factors in survival analysis, odds ratios and logistic regression. There is also a chapter on writing a statistical report which students would find more useful than others.

There is mention of statistical packages that implement certain techniques and, from a personal viewpoint, it would have been nice to see actual syntax from SAS, say. There were also a couple of typographical errors but this should not deter readers. Nevertheless, this is an excellent book and definitely recommended for those trying to learn about the area.

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Measurement, Design and Analysis: an Integrated Approach

E. J. PEDHAZUR AND L. PEDHAZUR SCHMELKIN, 1991

Hillsdale, Erlbaum

ISBN 0 8058 1063 3

The prospect of reviewing a book of over 800 pages is rather daunting but the title of this book certainly encouraged me to do so. Strictly I think that the volume should be subtitled 'A cross-referenced approach' (rather than integrated) as I do not really feel that the authors have persuaded the readership to think about these different issues simultaneously (and in a truly integrated fashion). After a quick overview, the book is divided into three sections, each of several chapters, followed by a list of references (48 pages!), tables of F and χ^2 (but only these tables), and detailed author and subject indexes.

Part 1 ('Measurement') is certainly worth a wide readership. It covers a variety of topics including scales of measurement, validation, reliability etc. in a deceptively simple manner but with some very critical discussion. I fear that some novices may give up on this section, considering it too pedestrian to waste their time on. But wide readership of this type of material would do far more for the quality of research than any number of books on the t -test. Forward reference to Part 3 is a little frustrating at times but well worth it to separate the concepts from the detailed methods. The exception is Chapter 5 on reliability which contains a fair amount of algebra as well as conceptual ideas.

Part 2 ('Design') does not contain the sort of stuff that I was hoping to see. There is very little of the type of material that traditionally comes under a heading of experimental design. Instead, much of the material might come under a heading of interpreting conclusions based on the way that data were collected. Chapter 11, for example, is headed 'Artefacts and pitfalls in research' and discusses the inadvertent influence that researchers and their subjects may have on conclusions. Chapter 12, specifically titled 'Experimental designs', discusses different macro concepts of designs—laboratory *versus* field experiments, for example—making sure that the 'treatment' is given in a pure and uniform manner, and other issues too. This is important but I did not see much on blocking, factorial experiments etc., a little, but not much. Randomization is stressed very early in this chapter as an important requirement for a true experiment but not much more is said about it subsequently. There is still little 'integration' of sections although cross-referencing does exist: but cross-referencing merely to Part 3 (all 400 pages of it) is not particularly helpful. The following chapter is headed 'Quasi-experimental designs'. I am not familiar with this term but (not surprisingly) it is concerned with observational studies. Randomization is mentioned again, the point being made that quasi-experiments do not have any but they should have all the other good features of experiments. This is then followed by a chapter on 'Non-experimental designs' which is perhaps a derogatory term for observational studies (I am sure that my epidemiologist friends will not like it).

Finally, we come to Part 3 ('Analysis'). As previously noted there are 400 pages of this so it ought to stand up against some of the heavier competition. The opening chapter is, interestingly, on software. The authors are appropriately supportive, yet critical. Many of us may be forced into a specific type of analysis because of limitations (generally, or locally) of software but, as the book points out, we should not fall into the trap of 'software-driven thinking'. The first 'methods' chapter is about simple linear regression. The data used are artificial; the mathematics are followed through step by step. We obtain estimates and partition out the sums of squares due to regression and residuals to obtain an

F-test. There is comment on how to do this in MINITAB and how to report the results. We get into fixed and random variables, residuals and diagnostics, outliers, leverage and more. Might this be information overload? The following chapters are on multiple regression, analysis of covariance (under different titles), factor analysis and structural equation models. Given the very introductory nature of some of the other material this seems to assume that much of the basics are already well grasped. Reference to software is widespread with small pieces of example code from MINITAB, SAS and SPSS. Each chapter also has a few questions (and answers) but these are really to test mathematical correctness rather than conceptual understanding. I do not like questions like ‘Given that $r_{y1} = .5$, $r_{y2} = .0$, $r_{12} = .3$, what is $r_{y(2.1)}$ equal to?’. The answer gives an equation number to refer back to so we could read around that section for explanation of what it all means but, given the emphasis on computers to do the work, the interpretation is more important, surely.

So who is the book for? Nowhere is it explicitly stated, although it soon becomes clear, that social scientists are probably the key audience. I fear that the authors have tried to do too much. The effort must have been considerable but all they have achieved over three separate (possibly already available) texts is, supposedly, ‘integration’. But here I think that they have failed. There is some (but not huge amounts of) cross-referencing but that is a second best to integration. Finally, this is supposed to be a ‘student edition’ but students would be better directed at small sections from many texts rather than trying to obtain it all from one.

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Applied Factor Analysis in the Natural Sciences

R. A. REYMENT AND K. G. JÖRESKOG, 1993

Cambridge, Cambridge University Press

xii + 372 pp.

ISBN 0 521 41242 0

This is the second edition of a book which originally appeared in 1976. Its first edition was produced by a different publisher, with a different title and a slightly different set of authors, and was reviewed in *Applied Statistics*, in 1977, by C. F. Banfield. The change of publisher has improved the appearance of the book, but I was surprised to find how little of the actual text has changed.

The title, *Applied Factor Analysis in the Natural Sciences*, is meant to indicate that this new edition has a wider scope than the original edition which was called *Geological Factor Analysis*. However, the vast majority of the examples still refer to geology and related areas and often use jargon that only someone with a geological background would fully understand. Although the preface to the first edition talks of a post-graduate course on the subject, the book looks more like a reference than a text-book—for example, there are no exercises.

It should be pointed out immediately that ‘factor analysis’ (FA) is used in a broad sense, akin to the term *analyse factorielle*, rather than in the more specific sense that is familiar to most statisticians. The authors state in the preface that the book is intended for natural scientists and that they have ‘definitely not written a text for statisticians’ but I suspect that it will not only be statisticians who find this deliberately chosen use of the term ‘factor analysis’ a little confusing.

The book consists of eight chapters and an appendix, which will now be described in turn, highlighting changes from the first edition. Chapters 1 and 2 are introductory and largely unchanged from the first edition. The first gives a motivating geological example in which a rotated principal component analysis (PCA) is discussed, followed by a large number of single-paragraph descriptions of other applications. Chapter 2 is a thorough review of mathematical and statistical concepts used later, concentrating on matrix algebra. There are some useful insights in this chapter, but any mathematically naïve readers would quickly become lost.

Both Chapter 3 and the first three sections of Chapter 4 are concerned with PCA and ‘true’ FA, and are, again, virtually unchanged from the first edition. There are useful reviews of standard properties of PCA and several varieties of FA. Differences are noted between PCA and FA, but PCA is nevertheless rather artificially forced into a ‘factor model’. The last four sections of Chapter 4 are new, dealing with ‘Robust principal component analysis’, ‘Principal component analysis and cross-validation’, ‘Principal component analysis of compositional data’ and ‘Path analysis and Wrightian factor analysis’. These